

Introduction to Control Systems (ME2801)

Homework #7: Frequency Response

HW7.1 Ratio → Decibels

The magnitude (AKA gain) of the transfer function is often expressed in decibels (dB) in the frequency response. Express the following gain values in decibels:

1. $|G| = 1$
2. $|G| = 2$, $|G| = 0.5$
3. $|G| = 10$, $|G| = 0.1$
4. $|G| = 4.2$, $|G| = 42$, $|G| = 420$, $|G| = 4200$

HW7.2 Decibels → Ratio

As an example, consider an unmanned surface vessel as shown in Figure 1.



Figure 1: Kingfisher USV

Consider the forward motion of this vehicle (the “surge” direction) where the input is a voltage to the electric motors (V_{in}) in Volts and the output is forward velocity ($\dot{x}_{out}$) in m/s – illustrated in Figure 2.



Figure 2: Input/Output of USV

Suppose we are given the magnitude portion of the frequency response (Bode plot) for this system in Figure 3.

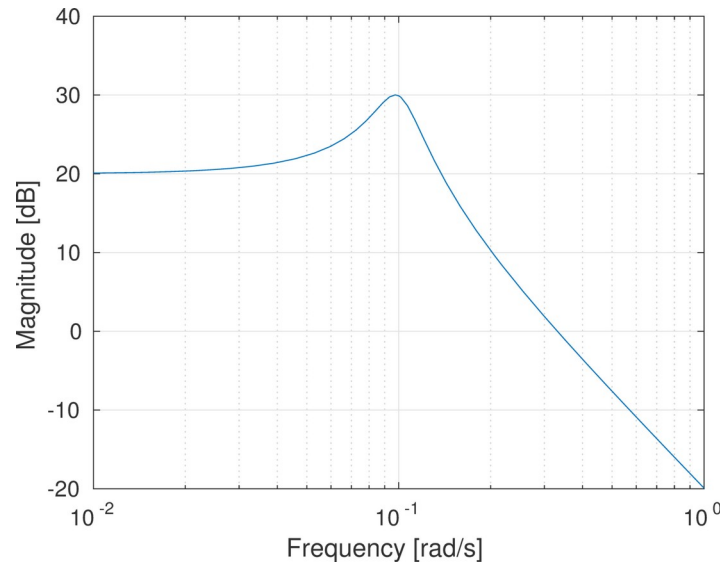


Figure 3: Magnitude portion of open-loop Bode plot for forward (surge) velocity response of USV.

Given the sinusoidal inputs below,

Find the steady-state output amplitude ($|\dot{x}_{out}|$) with appropriate units for each case:

1. V_{in} : Amplitude: 0.1 V ; Frequency: 0.01 rad/s.
2. V_{in} : Amplitude: 1.0 V ; Frequency: 0.1 rad/s.
3. V_{in} : Amplitude: 0.1 V ; Frequency: 1.0 rad/s.

HW7.3 Bode Plots: Asymptotic Approximations

Sketch the asymptotes of the Bode plot (magnitude and phase) for the following open-loop transfer functions. Submit your hand sketch and identify the slope values for the asymptotes and the key values of the magnitude and phase plots. You may use MATLAB to generate the Bode plots as a guide, but only submit your hand sketch.

Use the [Bode graph paper provided on the course website](#) for sketching the frequency response.

For the magnitude plot, include labels for any non-zero slope values

1. $L(s) = \frac{2000}{s+200}$

2. $L(s) = \frac{2000}{s(s+200)}$ (F&P 6.3)

3. $G(s) = \frac{0.1}{(s^2+s+4)}$

4. $G(s) = \frac{0.1(s+10)}{(s^2+s+4)}$

Note: You can use the results from part 1 to build the Bode plot for part 2. Similarly, you can use the result from part 3, to help generate part 4.